[2CEIT401 – Operating Systems]

Practical: 4

AIM- Perform various file operations.

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Aim: Perform various file operations.

1. Write a shell script to check whether the entered number is prime or not.

Ans:

Code:

```
echo "Enter Number: "
read num
i=2
isPrime=1
while [ $i -lt $num ]
do
   temp=`expr $num % $i`
   if [ $temp = 0 ]
   then
   isPrime=0
   fi
   i=`expr $i + 1`
done
if [ $isPrime = 1 ]
then
   echo "Number Is Prime"
else
   echo "Number Is Not Prime"
fi
```

Output:

```
[24012011142@linuxserv prac4]$ sh que_1.sh
Enter Number:
15
Number Is Not Prime
[24012011142@linuxserv prac4]$ sh que_1.sh
Enter Number:
7
Number Is Prime
```

2. Write a shell script to calculate HRA of employees depending upon their basic.

Ans:

Code:

```
echo "Enter Basic Salery: "
read sal
if [ $sal -lt 10000 ]
then
hra=`expr $sal \* 20 / 100`
hra_r=20
elif [ $sal -lt 20000 ]
then
hra=`expr $sal \* 25 / 100`
hra_r=25
else
hra=`expr $sal \* 30 / 100`
hra_r=30
fi

total=`expr $sal + $hra`

echo "Sal: $sal"
echo "HRA Rate %: $hra_r"
echo "HRA: $hra"
echo "Gross Salary: $total"
```

Output:

```
[24012011142@linuxserv prac4]$ sh que_2.sh
Enter Basic Salery:
8000
Sal: 8000
HRA Rate %: 20
HRA: 1600
Gross Salary: 9600
[24012011142@linuxserv prac4]$ sh que_2.sh
Enter Basic Salery:
15000
Sal: 15000
HRA Rate %: 25
HRA: 3750
Gross Salary: 18750
[24012011142@linuxserv prac4]$ sh que_2.sh
Enter Basic Salery:
40000
Sal: 40000
HRA Rate %: 30
HRA: 12000
Gross Salary: 52000
```

3. Write a shell script that greets the user by saying Good Morning, Good Afternoon, and Good Evening according to the system time.

Ans:

Code:

```
hour=`date +%k`
if [ $hour -ge 5 ] && [ $hour -le 12 ]
then
echo "Good Morning"
elif [ $hour -ge 13 ] && [ $hour -le 17 ]
then
echo "Good Afternoon"
elif [ $hour -ge 18 ] && [ $hour -le 20 ]
then
echo "Good Evening"
else
echo "Good Night"
fi
```

Output:

```
[24012011142@linuxserv prac4]$ date
Mon Feb 16 02:45:10 IST 2026
[24012011142@linuxserv prac4]$ sh que_3.sh
Good Night
```

4. Write a shell script that asks the capital of Gujarat and repeats the question until the user gives correct answer.

Ans:

Code:

```
isTrue=0

while [ $isTrue = 0 ]

do

echo "Enter Capital Of Gujarat:"

read cap

if [ $cap = "gandhinagar" ] || [ $cap = "Gandhinagar" ]

then

isTrue=1

fi

done
```

Output:

```
[24012011142@linuxserv prac4]$ sh que_4.sh
Enter Capital Of Gujarat:
gandhi
Enter Capital Of Gujarat:
Gandhinagar
[24012011142@linuxserv prac4]$
```

5. Write a shell script to display desired line from a file.

Code:

```
line=$1

file=$2

head -n $line $file | tail -n 1
```

Output:

```
[24012011142@linuxserv prac4]$ cat abc.txt
Hii My
My Name
Name is Vatsal
Is Vatsal
Vatsal, I am Doing
I
am
Doing
OS
Practical
[24012011142@linuxserv prac4]$ sh que_5.sh 5 abc.txt
Vatsal, I am Doing
[24012011142@linuxserv prac4]$
```

6. Write a shell script to count number of newline characters in a file.

Code:

```
file=$1

lines=`wc -l $file | awk '{ print $1 }'`

echo "Lines: $lines"
```

Outputt:

```
[24012011142@linuxserv prac4]$ sh que_6.sh abc.txt
Lines: 10
[24012011142@linuxserv prac4]$
```

7. Write a Shell script, which counts the number of words in a file, without taking into consideration the blank space, tab spaces and the newline characters using WC.

Code:

```
file=$1

words=`wc -w $file | awk '{ print $1 }'`

echo "Words: $words"
```

Output:

```
[24012011142@linuxserv prac4]$ sh que_7.sh abc.txt
Words: 18
[24012011142@linuxserv prac4]$
```

8. Write a Shell script, which counts the number of words in a file, without taking into consideration the blank space, tab spaces and the newline characters without using WC.

Code:

```
file=$1

words=`awk '{num+=NF} END{print num+0}' $file`

echo "Words: $words"
```

Output:

```
[24012011142@linuxserv prac4]$ sh que_8.sh abc.txt
Words: 18
[24012011142@linuxserv prac4]$
```